

Measuring the Efficacy of a Low-Cost Tendon-Driven Prosthetic Arm

Evaluating the dexterity, strength, and durability of prosthetics made with affordable materials



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The Problem: Rejection, Cost, and Rising Need

44%

Upper-limb amputees
rejecting their prosthesis
(Salminger *et al.*, 2022)

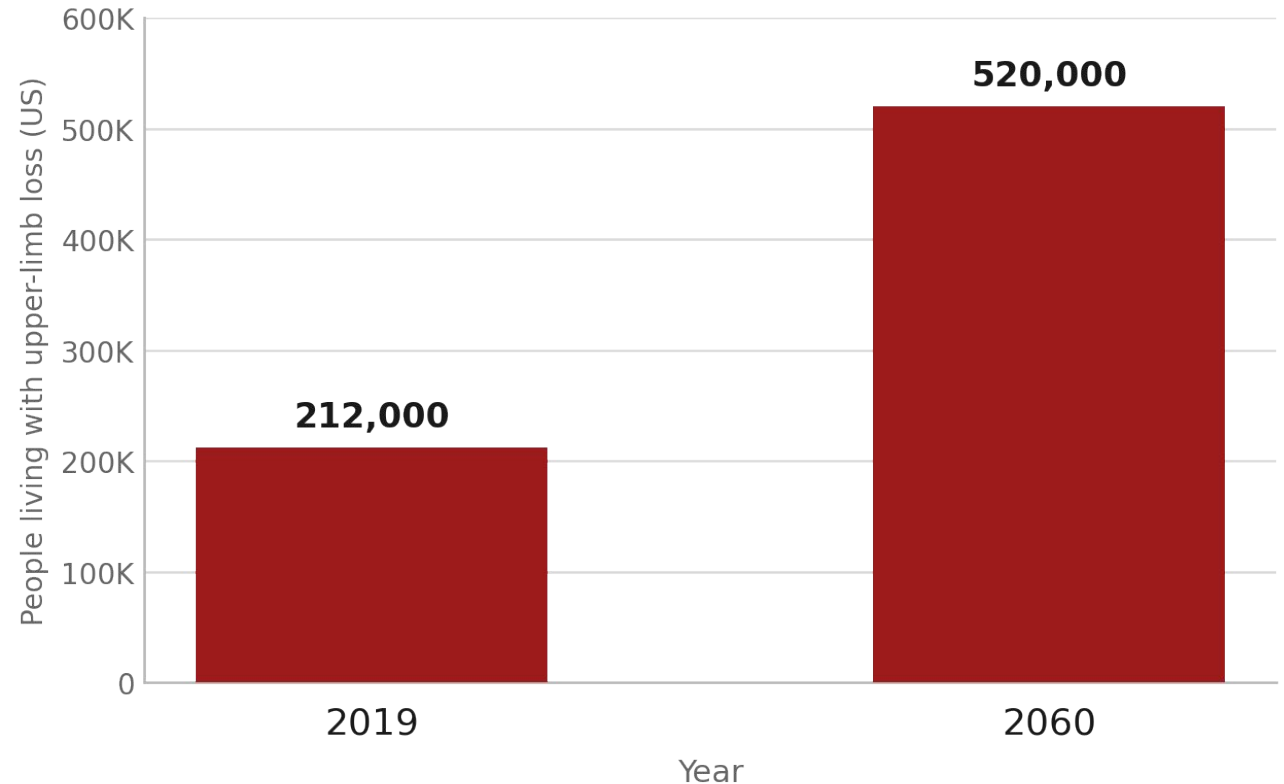
81%

of non-users citing
limited functionality
(Yamamoto *et al.*, 2019)

≈\$6,000

Price of Hero Arm, a
lower bound of
prosthetics cost
(Open Bionics, 2026)

Projected US upper-limb loss



(Rivera *et al.*, 2024)



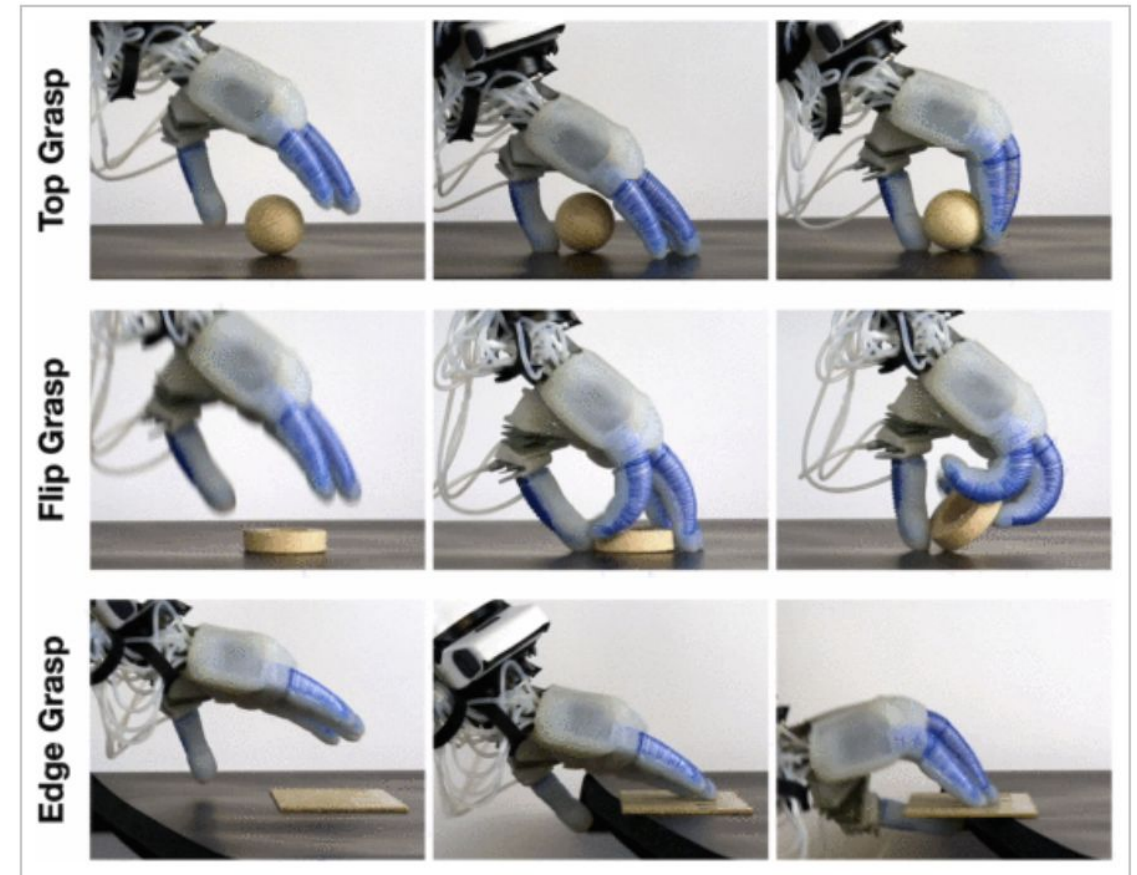
Mission Statement

Study the effect of utilizing cheap, hobbyist materials on prosthetic arms, specifically dexterity and torque output, which would contribute to the development of more affordable and accessible prosthetic through our research outcome.

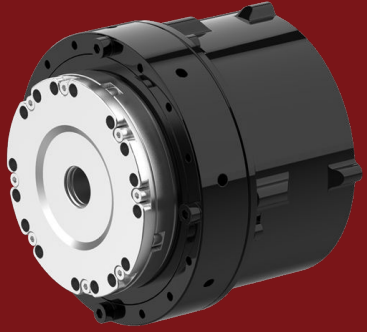


Literature Review: Benchmarks and Prior Work

- **Preset grips**
Commercial hands: typically have between 4 and 36 preset grip patterns (Open Bionics, 2023; Waryck, 2011)
- **GRASP taxonomy: A Universal Benchmark?**
33 grasp types: independent, universal dexterity standard (Feix *et al.*, 2016)
- **RBO Hand 3: An Example Case**
Soft robotic hand benchmarked on all 33 GRASP types (Puhlmann *et al.*, 2022)
- **Why users abandon devices**
Weight, function, cost, unchanged acceptance since 2006 (Henao *et al.*, 2026; Salminger *et al.*, 2022)



RBO Hand 3 Performing Grasps
Note. From Puhlmann *et al.* (2022).



Primary Research Hypothesis

A low-cost prosthetic arm can be produced while meeting the following criteria:



< \$300

per unit cost

$\geq 18 / 33$

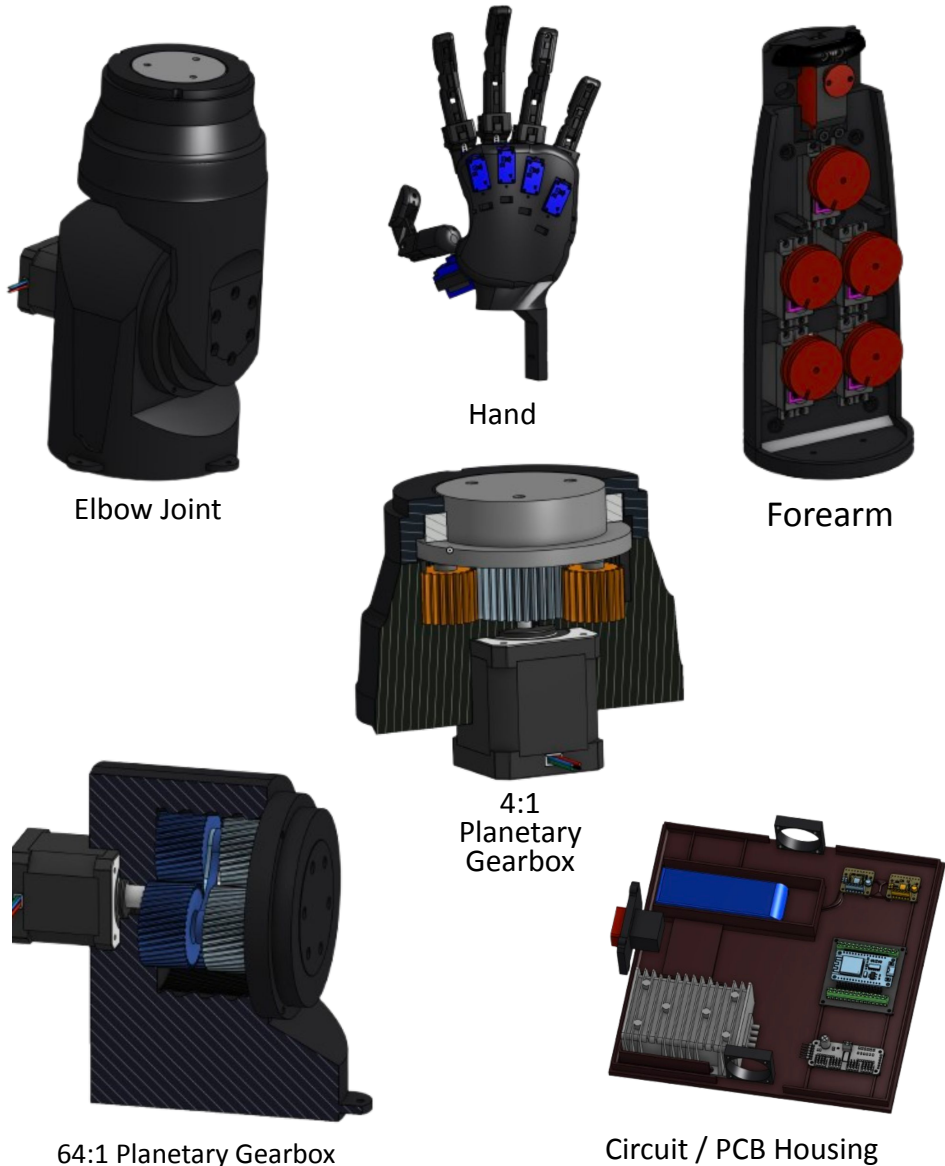
GRASP types

$\geq 10 \text{ N}\cdot\text{m}$

elbow torque

Benchmarked against the GRASP taxonomy by Feix *et al.* (2016).

Methods: “ARMin” System Architecture



\$290.54 cost per finished unit

Actuation

- 6 servos
- 5 micro servos
- 2 stepper motors

Hand

- Fishing-line tendons
- Elastic return
- Direct-drive splaying

2-DOF Elbow

- Planetary actuators
- 64:1 + 4:1 reductions

Control

Computer Vision → ESP32
→ arm

Structure

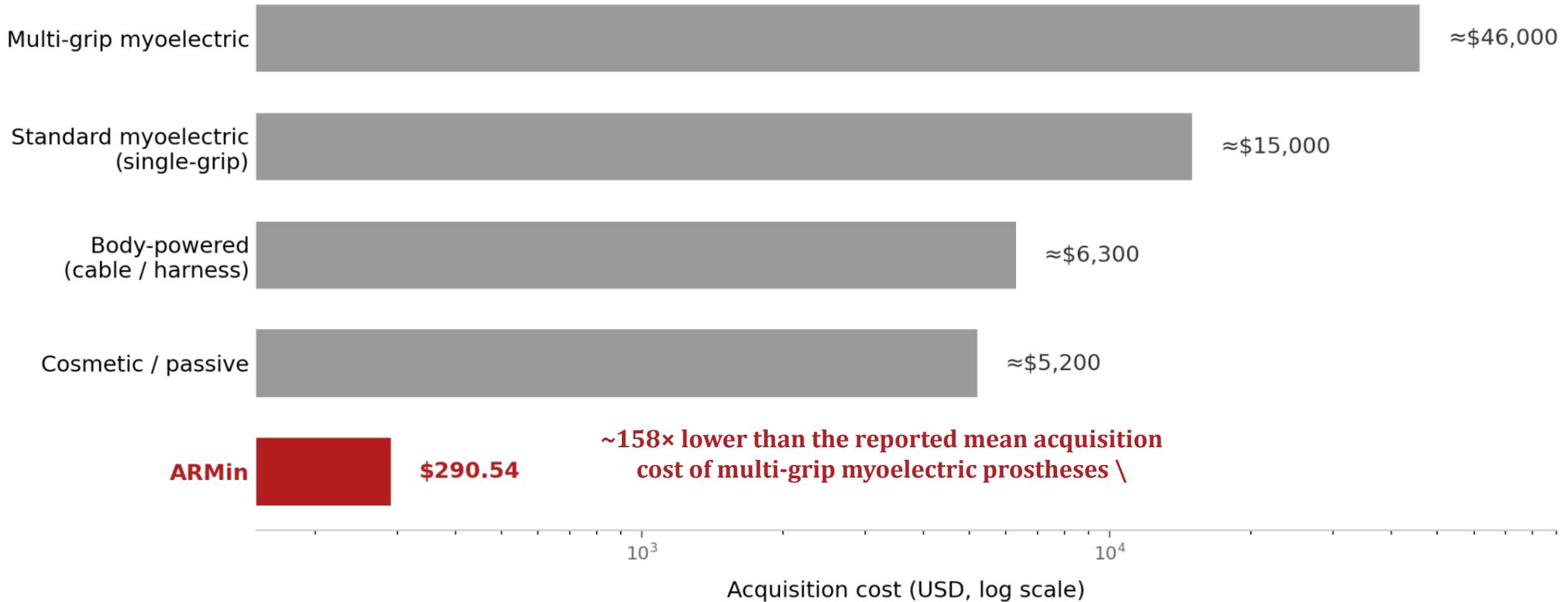
- Fully 3D-printed
- PLA + PETG

Power

4S LiPo → buck converter
→ terminal block →
PCA9685 PWM driver

Cost Comparison and Budget

What a comparable arm costs



Commercial figures: mean acquisition costs by prosthesis class (Kerver *et al.*, 2023), converted from EUR to USD; approximate. ARMin: measured bill of materials.

Grant Request

Equipment needed for arm classification and reproducible data collection

Equipment	Purpose	Min	Max
S-type tension load cell, 0–500 N, with digital indicator	Payload and failure load at the elbow	\$90	\$210
Test fixture — aluminum extrusion frame, machinist vise	Fixed moment arm across all trials	\$120	\$260
Cycling rig — motorized cam/crank, cycle counter	Automated open–close cycles for tendon durability	\$120	\$300
Calibration check mass — cast-iron grip weight	Calibration of load cell	\$40	\$110
Trial documentation — 2× 1080p60 cameras, tripods, LED panel	Documentation	\$150	\$320
Second ARMin arm — duplicate of current build, per BOM	Test unit for dexterity, torque, and durability (cycled to failure)	\$290	\$290
Total request		\$810	\$1,490

Load applied at a fixed moment arm so force (N) converts to a reproducible elbow torque (N·m). The cycling rig runs the 10,000-cycle tendon-durability test; the second arm is cycled and loaded to failure while the current build stays intact as a working demonstrator.

Ranges are current retail estimates and set the minimum and maximum request.

Experimental Design: Variables and Measurements

Independent variable

- Grasp type
- Applied load
- # of Open-Close cycles

Dependent variables

- Grasp Success Rate
- Elbow Torque (N·m)
- Cycles till Failure

1

GRASP taxonomy assessment

- All 33 grasps
- Fixed test environment
- Dexterity score

Feix et al., 2016

2

Elbow payload

- Elbow flexion loaded at a fixed moment arm
- Load cell measures force output in Newtons

3

Tendon durability

- Motorized cycling rig drives repeated 10k open–close actuation
- Inline load cell logs peak tendon tension each cycle
- Failure mode recorded: stretch, fray, or slip

Possible Outcomes: Three Hypothesis Variations

Primary Hypothesis

18/33 GRASP types or more
exhibit passing criteria

AND

Sustained 10 N·m elbow torque

Alternative Hypothesis

18/33 GRASP types DO NOT
exhibit passing criteria

OR

Sustained 10 N·m elbow torque

Null Hypothesis

18/33 GRASP types DO NOT
exhibit passing criteria

AND

Inconsistent 10 N·m elbow
torque

Grounding for expected performance

- 33-grasp benchmarking demonstrated on anthropomorphic hands (Puhlmann *et al.*, 2022)
- Preset-grip devices → functional abandonment (Yamamoto *et al.*, 2019; Biddiss & Chau, 2007)

Future Research

1 Fully open-source release

- CAD
- Firmware
- Build documentation

2 Brain–computer interface control

Neural motor prostheses path

(Hochberg *et al.*, 2006; Arshad *et al.*, 2022)

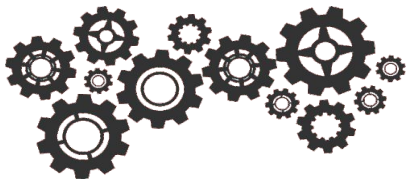
3 Mounting harness

Designing a socket interface for amputee testing

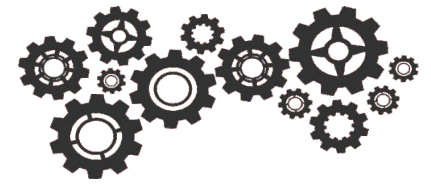
4 Benchmark profile

- Head-to-head vs. industry standards on GRASP
- Torque
- Durability

We believe this research should proceed because a sub-\$300, open-source arm that meets objective dexterity and strength benchmarks would give amputees without access to care a functional alternative that commercial pricing currently denies them.



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Demonstration!

Thank You!

Questions?